

Srijan Basu

Full Stack Developer | Backend-Leaning | ML Engineering

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GitHub | LinkedIn | Portfolio

SUMMARY

Fresher Full Stack Developer with 6+ months of hands-on internship experience building production-grade backend systems at an early-stage healthcare AI startup. Specialized in FastAPI, PostgreSQL, and multi-tenant SaaS architecture. Comfortable across the stack — from REST API design and JWT authentication to React dashboards, ML model deployment, and cloud deployments. Seeking backend, full stack, or ML engineering roles where I can ship real features and grow fast.

TECHNICAL SKILLS

- **Backend & APIs:** Python, FastAPI, Node.js, Express.js, TypeScript, RESTful API design, layered/service architecture
- **Databases:** PostgreSQL, MongoDB, DynamoDB (TTL-based OTP), SQLAlchemy ORM, relational schema design
- **Machine Learning:** Scikit-learn, Pandas, NumPy, Random Forest, model serialization (Joblib), ML pipeline design, feature engineering, model deployment
- **Data Visualization:** Plotly, Streamlit, interactive dashboard development
- **Auth & Security:** JWT (HS256), bcrypt, OTP authentication, role-based access control (RBAC)
- **DevOps & Tools:** Git/GitHub, Swagger UI, Postman, Vercel, Netlify, Streamlit Cloud
- **Generative AI:** Gemini API (LLM integration), prompt engineering, AI-powered feature development
- **Frontend:** React (v19), Next.js, Tailwind CSS, Vite, custom hooks, Context API
- **Also familiar with:** C++, Django

WORK EXPERIENCE

- **Callsure.ai** (Early-stage voice-first healthcare AI startup) Sep 2025 – Present
Full Stack Developer Intern *Remote*
 - Architected multi-tenant healthcare SaaS backend with PostgreSQL relational models (**Hospital**, **User**, **Doctor**, **Appointment**), using `hospital_id` foreign key isolation and SQLAlchemy ORM with indexed queries to support concurrent multi-hospital operations
 - Engineered automated user onboarding pipeline: PostgreSQL insertion → SMTP transactional email (role-specific HTML/plaintext templates) → JWT token generation (7-day expiry, HS256, bcrypt hashing), enabling passwordless first-login flows for new staff
 - Built hybrid authentication system with bcrypt password validation and DynamoDB TTL-based OTP (10-minute auto-expiry) via FastAPI dependency injection (`get_current_user`), securing 30+ role-protected endpoints across 4 user roles (owner/admin/doctor/staff)
 - Designed service-layer FastAPI architecture separating route controllers (`routes/auth.py`, `routes/appointment.py`) from business logic services (`AppointmentService`, `DoctorService`), improving code maintainability and testability
 - Developed React 19 admin/staff dashboard with Vite: modular components, custom data-fetching hooks (`useAppointments.js`, `useDoctors.js`), JWT-based `AuthContext`, and responsive Tailwind CSS layouts
- **LTIMindtree** (Fortune 500 IT services company) Jun 2025 – Aug 2025
Cloud Infrastructure Service Engineer *On-Site, Kolkata*
 - Managed enterprise IT operations using WSUS, MECM, and Active Directory for user provisioning and patch management across a large-scale corporate environment

- Supported cloud infrastructure workflows and ITSM processes, gaining exposure to enterprise-grade tooling including OKTA (SSO) and Microsoft 365 administration

PROJECTS

- **Life Expectancy Intelligence Dashboard | Live Demo | GitHub** 2026
ML | Python, Scikit-learn, Pandas, NumPy, Streamlit, Plotly, Joblib, Git, Streamlit Cloud
 - Built a complete end-to-end ML pipeline on WHO health data (2,938 records, 193 countries, 2000–2015): data cleaning, feature engineering, model training, serialization, and cloud deployment
 - Trained a Random Forest Regressor (100 trees, Scikit-learn) achieving $R^2 = 96.91\%$ and $MAE = \pm 1.05$ years; persisted model with Joblib and enforced feature column order via JSON to prevent silent inference errors
 - Developed a 5-tab interactive Streamlit dashboard: country explorer with radar chart vs global average, real-time AI predictor (18 sliders), what-if policy simulator with per-lever marginal impact breakdown, feature importance visualization, and a plain-English data query engine built on pure Pandas — no external AI API
 - Deployed on Streamlit Community Cloud via GitHub with cold-start auto-training: model rebuilds itself on the server if `.pkl` not found, eliminating manual redeployment
- **Resumware – AI-Powered ATS Resume Builder | Live Demo | GitHub** Feb 2025 – May 2025
Full Stack | React, Node.js, Express.js, TypeScript, MongoDB, Gemini API, LaTeX
 - Built a full stack AI-powered resume builder that generates ATS-optimized resumes using LaTeX templates and LLM-based content generation via Gemini 1.5 Flash API
 - Designed and implemented REST APIs using Node.js, Express.js, and TypeScript; stored resume data in MongoDB; deployed on Netlify and Vercel with Git-based CI/CD workflows
- **Daily Finance Collection App** May 2024 – Jul 2024
Self Project | Django, HTML, CSS, JavaScript
 - Built a daily income/expense tracking app using Django; adopted by local small businesses to manage and summarize daily financial transactions

EDUCATION

- **Brainware University** Kolkata, India
Master of Computer Applications (MCA) — CGPA: 8.38 Sep 2025 – 2027 (Expected)
- **Techno India University, Salt Lake** Kolkata, India
Bachelor of Computer Applications (BCA Honors) — CGPA: 7.47 Sep 2022 – May 2025

CERTIFICATIONS

- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate** — Oracle (Oct 2025, valid until Oct 2027) | View Certificate
- **AI-ML Virtual Internship (10 weeks)** — AICTE–EduSkills (Google for Developers), Jul–Sep 2024 | View Certificate